Sathwik Bollepalli

860-781-1558 | sathwik238@gmail.com | LinkedIn | Portfolio | GitHub

SUMMARY

Data Scientist skilled in predictive modeling, machine learning, and data visualization, with a track record of improving model accuracy by 15% and reducing processing time by 70%. Proficient in Python, SQL, and Tableau for delivering actionable insights. Certified in Modern **Big Data Analysis with SQL**, **Google Data Analytics**, Hacker Rank Python & Advanced SQL, and JPMorgan Chase CADP.

PROFESSIONAL EXPERIENCE

Kemper Insurance – Remote, CT, USA

Associate Data Scientist April 2025 – Present

- **Delivered an end-to-end, production-ready CV Underwriting Decision Support (UDS) pipeline**, using 10 years of commercial auto data and Snowpark ETL to standardize **POLICY, XP, CLM, INC, and FIN** roll-ups; built scalable **feature-engineering modules**, developed and tuned the frequency model, performed rigorous **model validation** (lift, loss-ratio, predicted-distribution stability), and implemented automated **risk-scoring** logic for real-time underwriting deployment.
- Developed Personal Auto **Pure Premium pricing models** for COLLISION coverage using H2O GLM (Poisson/Tweedie + B-splines), improving rate adequacy by running variable selection, lift analysis, and diagnostics across statewide rating factors. Enhanced risk assessment by applying localized adjustment factors.
- Built an automated **Snowflake Data Profiler** (HTML stats, distributions, charts) and an **LLM-driven Schema Analyzer** to classify variables and surface modeling insights using only metadata.
- Researched **BitNet 1.58** quantized LLMs, benchmarking memory savings and inference trade-offs, and delivered guidance for potential low-memory AI deployment in pricing and UDS tooling.
- Built a **Python package** to manually compute underwriting factor tables (territory, prior-carrier, selection-score), enabling precise **model-to-production reconciliation** and significantly improving **post-launch validation of the Pure Premium model** by identifying inconsistencies in deployed rating factors.

Plymouth Rock Assurance - Boston, USA

Data Scientist Intern May 2024 – April 2025

- Achieved a **70% reduction in data processing time** by restructuring pipelines, using data structures and automating the generation of 800+ features, enabling the scalability of predictive models and reducing deployment delays.
- Developed Tweedie XGBoost models with Bayesian hyperparameter tuning, achieving a 15% improvement in Gini coefficient and more accurate home insurance premium predictions.
- Enhanced risk assessment by applying localized adjustment factors, increasing prediction accuracy for high-risk areas like flood zones and high-crime neighborhoods.
- Improved precision for predicting large-loss fire events by **30%** in highly imbalanced datasets, leveraging advanced resampling techniques and XGBoost-based logistic regression.
- Influenced key strategic decisions by delivering insights to senior management, resulting in the integration of two high-value third-party data sources, improving model performance and decision-making processes.

MAQ Software Pvt Ltd - Hyderabad, India

Machine Learning Engineer *May 2022 – August 2023*

- Achieved a 50% improvement in data retrieval efficiency and **reduced processing time by 77%** (from 2 hours to 27 minutes), directly enhancing operational workflows and enabling faster decision-making across the organization.
- Improved model deployment efficiency through the implementation of a no-code approach in Azure ML, reducing manual errors and accelerating the time-to-production for machine learning models.
- Automated data extraction and organization processes, resulting in a **40% reduction in data preparation time** and increasing data accessibility for analytics teams, improving workflow reliability.
- Increased **stakeholder engagement by 35%** and improved decision-making by providing clear, actionable insights through visually compelling dashboards that synthesized complex data into easily digestible formats.
- Enhanced analytics and reporting accuracy by 20%, enabling the delivery of more precise insights to stakeholders and reducing data inconsistencies across reporting platforms.

EDUCATION

University of Connecticut – Storrs, CT August 2023 - December 2024

Master of Science (M.S) in Data Science, ***GPA : 3.7***

Relevant Courses – Predictive Modeling, Machine Learning, Statistical Analysis, Causal Inference, Data Visualization, Deep Learning.

TECHNICAL SKILLS

Python (Pandas, NumPy, PySpark, Scikit-learn, XGBoost, LightGBM, CatBoost, Matplotlib, Seaborn, Keras, TensorFlow), R, SQL (MySQL, PostgreSQL, Snowflake), H2O GLM (Poisson/Tweedie/NB), pure-premium and frequency/severity modeling, loss-cost modeling, A/B testing, hypothesis testing, Bayesian analysis, ETL pipelines with Snowpark/Snowflake, AWS (S3, EC2, SageMaker), Azure (Data Factory, Databricks), deep learning (CNN, RNN, LSTM), transformers (BERT, GPT), LLM/quantization research (BitNet 1.58), feature engineering, predictive modeling, clustering, data wrangling, automated data profiling, Tableau, Power BI, Git, Excel.